



Impact of misinformation from generative AI on user information processing: How people understand misinformation from generative AI

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Donghee Shin 
Texas Tech University, USA

Amy Koerber
Texas Tech University, USA

Joon Soo Lim 
Syracuse University, USA

Abstract

This study examines the impact of artificial intelligence (AI) on the ways in which users process and respond to misinformation in generative artificial intelligence (GenAI) contexts. Drawing on the heuristic–systematic model and the concept of diagnosticity, our approach examines a cognitive model for processing misinformation in GenAI. The study’s findings revealed that users with a high-heuristic processing mechanism, which affects positive diagnostic perception, were more likely to proactively discern misinformation than users with low-heuristic processing and low-perceived diagnosticity. When exposed to misinformation from GenAI, users’ perceived diagnosticity of misinformation can be accurately predicted by the ways in which they perform heuristic systematic evaluations. With this focus on misinformation processing, this study provides theoretical insights and relevant recommendations for firms to be more resilient in protecting users from the detrimental impacts of misinformation.

Corresponding author:

Donghee Shin, Digital Media & Professional Communication, College of Media & Communication, Texas Tech University, 3003 15th St, Lubbock, TX 79409, USA.

Email: don.h.shin@ttu.edu

Keywords

Algorithmic effects on misinformation, algorithmic misinformation, ChatGPT, generative AI, heuristic–systematic process, misinformation-processing model

Introduction

ChatGPT, a chatbot leveraging human-like artificial intelligence (AI) and generative AI (GenAI) capabilities, has developed as part of the Chat Generative Pretraining Transformer (ChatGPT) series and rapidly gained popularity as a versatile tool for information retrieval and various generative tasks. This viral AI-powered chatbot can write essays, theses, visuals, and even computer code. ChatGPT's dialog format allows it to answer follow-up questions, admit mistakes, challenge incorrect premises, and make appropriate suggestions (Brynjolfsson et al., 2023).

While AI-generated text has gained immense popularity, experts have expressed concern that its unverified use could lead to the widespread dissemination of misinformation masquerading as credible scientific knowledge (Eysenbach, 2023; Van Dis et al., 2023). Verifying the accuracy of GenAI-generated content is a substantial challenge due to inherent limitations in AI models, the opaqueness of their algorithms, and the potential for contamination in their data sources (Shin and Valente, 2020; UK National Cyber Security Center, 2023). Furthermore, GenAI's inherent tendency to generate plausible-sounding misinformation can exacerbate the challenge of verifying GenAI-generated content (Brynjolfsson et al., 2023).

To better understand user perceptions of potential misinformation in AI systems and their challenges in verification, it is essential to understand the source and nature of misinformation generated by ChatGPT. Misinformation is defined as false or objectively incorrect information that is not sustained by expert opinion or evident proof (Vraga and Bode, 2020). Misinformation-related problems are exacerbated by the wide availability of online platforms and the instantaneous and viral nature of online discussions (Margolin, 2021). With the advent of GenAI, the misinformation dilemma has escalated. ChatGPT has attracted enormous traction due to its ability to present information quickly and confidently to users (UK National Cyber Security Center, 2023). However, this speed and capability can pose risks, as ChatGPT frequently generates false narratives, particularly in health misinformation (Walter et al., 2021). The efficient and quick generation of information by GenAI could have wide-ranging implications for the diffusion of health misinformation online (Zhao and Tsang, 2023). For example, ChatGPT cannot discern the boundary between fact and likelihood (Van Dis et al., 2023). Owing to this inability, ChatGPT tends to fabricate information to satisfy users' inquiries with a quick response time. Since misinformation can flow into AI models as well as from them, some GenAIs can become vulnerable to injection attacks. In these attacks, malicious users intentionally input false information into the models during training, causing the AI models to propagate these falsehoods when generating outputs (UK National Cyber Security Center, 2023). ChatGPT is especially prone to being exploited for health-related misinformation generation and amplification (Eysenbach, 2023). In many cases, it presents health misinformation as fact and truth (Melchior and Oliveira, 2022; Zhao and Tsang, 2023). The

natural language format of ChatGPT's responses can lead users to trust algorithmically generated information. This concern is related to a lack of transparency and explainability (Shin, 2023). For ChatGPT-generated answers, explaining to users where the information comes from and how the answers are generated, constructed and presented to them remains a challenge. End users cannot and, in general, do not attempt to construe a sense of transparency and explainability in GenAI, which ultimately hurts user trust and engagement (Shin and Valente, 2020).

Another issue with GenAI is that the information it generates is disingenuous, as it reflects not the original thoughts or beliefs of a human but rather patterns in the text on which it was trained (UK National Cyber Security Center, 2023). This issue has been documented in several studies; for instance, Tamkin et al. (2021) discussed the potential for large language models, like GPT-3, to be misused for generating disinformation in various forms, including false or misleading essays, tweets, and news stories. De Angelis et al. (2023) highlighted the significant use of ChatGPT in composing parts of scientific articles and reports, cautioning against the potential spread of misinformation. They pointed out the risk of creating "seemingly legitimate but actually fabricated or misleading content" (De Angelis et al., 2023: 4), leading to an "AI-driven infodemic" (p. 5), a novel public health threat. In addition, a lack of transparency about the specifics of its pre-training data and supervised fine-tuning tasks adds to the challenge. ChatGPT does not offer explainability for its output generation process nor discloses its uncertainty level. Consequently, when dealing with topics outside its adequately trained scope, ChatGPT might produce fabricated responses with unwarranted certainty, a phenomenon termed "artificial hallucination" (Varghese and Chapiro, 2023). ChatGPT does not always grasp the full context of a subject, which can cause it to generate overly literal or nonsensical answers. Eysenbach (2023) noted another limitation of the large language models, such as its tendency to hallucinate or fabricate information, including references, and its dependence on precise prompting for accurate responses. This limitation stems from the inherently non-human nature of these AI models. As Mishra et al. (2023) explain, humans are social beings whose cognition significantly involves understanding social norms, beliefs, and desires. In contrast, AI models like ChatGPT lack personal experiences, emotions, or consciousness, operating solely on learned patterns without any emotional depth or personal insight.

GenAI, including ChatGPT, often struggles with highly specialized or politically sensitive subjects, especially when trained on data that may not be comprehensive or accurate. For example, when asked about cutting-edge scholarly medical information, ChatGPT can mislead users by generating easily available information online. Also, when asked about sensitive issues such as LGBTQ+ or abortion, GenAI might reflect the implicit bias of the content creators by providing subjective opinions or might overlook the opinions or knowledge of marginalized communities. GenAI will not only inundate the Internet with more misinformation but will also generate convincing disinformation that is targeted at vulnerable users (Brynjolfsson et al., 2023). Thus, deciding what is "true" is a matter of politics and perspective. Given this, the challenge of addressing misinformation can become complicated depending on what information an AI is trained on and who creates its platforms and algorithms. Despite ongoing efforts to overcome these potential dangers of GenAI tools, designing a system in which AI

generates accurate information remains a major challenge, particularly in health contexts (Shin and Valente, 2020). Therefore, it is important to continue researching how human users can become more effective at identifying AI-generated misinformation in health contexts. Understanding users' heuristics and sense-making models about the roles of cognitive processes when interpreting health misinformation can help combat misinformation and the spread of fake news in the GenAI landscape (Pennycook and Rand, 2022; Wu et al., 2023).

Modeling the perceptual and cognitive processes behind the spread of misinformation is pertinent to understanding users' reasoning when encountering misinformation. Along these lines, numerous theories (e.g. debunking, informative fiction, and inoculation theory) from various fields have been formalized into models that attempt to analyze how an individual user responds to misinformation (e.g. Borukhson et al., 2022). Many methods for combating misinformation have been explored, and studies have investigated how people encounter and share false information or misinformation online (e.g. Bryanov et al., 2020; Vu and Chen, 2023). Among all attempts at understanding misinformation, efforts should focus on the processes that can help users to more critically engage with or more consciously respond to misinformation (Molina and Sundar, 2023). Although the users' cognitive processing mechanisms of misinformation are important, particularly in GenAI contexts, they have remained under-researched due to the nascent stages of the technology. There is burgeoning research on how the accuracy of misinformation is automatically determined by AI. For example, the utility and efficacy of interventions of fact-checking practices (Lee et al., 2023) and technical tools, such as automated fact-checking (Epstein et al., 2021), prebunking (Lewandowsky and Van der Linden, 2021), and debunking (Ecker et al., 2022) have been studied extensively. However, research focusing on users' cognitive heuristics for diagnosing misinformation, the related processes for discerning misinformation, and the ensuing effects of misinformation–amplification behavior is limited.

Filling this research gap requires better knowledge of the mechanisms of the relationship between users' misinformation-processing behavior and their cognitive dimension and development in GenAI (Walter and Tukachinsky, 2020). We narrow this research gap by examining how users respond to and process health misinformation in GenAI contexts. Because a great deal of today's misinformation is generated and amplified by AI (Peng and Wang, 2023), understanding how users process misinformation could help advance the theoretical foundations and practical solutions for the many problems caused by the spread of misinformation. This study's findings contribute to the ongoing literature by clarifying the algorithmic effects of AI constraints (Peng and Wang, 2023; Shin and Valente, 2020), such as the black-box nature and procedural uncertainty of AI, on users' evaluations of generated misinformation.

Literature review: the sense-making of generative misinformation

To examine the cognitive processes that lie behind users' responses to misinformation, we integrated insights from two theoretical accounts—the heuristic–systematic (HS) processing framework and the concept of information diagnosticity—and examined the

antecedents and consequences of misinformation processing by GenAI users. HS processing is particularly useful in health contexts as users are involved with the assessment of diagnostic information.

HS misinformation processing

The HS model of information processing analyzes the attitude formation and cognitive processing of individuals according to two types of response to information: heuristic and systematic (Chaiken et al., 1996). In heuristic responses, information is processed intuitively and autonomously, based on available heuristic cues or simple decision rules, with little reflection on the content of the message. In systematic responses, by contrast, users thoroughly evaluate the information to check its validity, and they reflect on the content of the message (Todorov et al., 2002). Both processes can be influenced by contextual factors. For example, when a topic is personally relevant to a user and their motivation to process accurately is high, they will be more likely to use systematic cues to process information. Thus, someone with high prior knowledge about a topic and the motivation to process information will process in accordance with both systematic and heuristic cues. The cooccurrence of both heuristic and systematic processing (users use both processing) in specific contexts (e.g. misinformation and AI) can be highly probable.

This theory is used as a lens for examining users' sense-making of misinformation; it clarifies the role of algorithmic features in users' sense-making as it relates to algorithms and how the actions of algorithms influence this sense-making (Zrnec et al., 2022). By examining the antecedents of user heuristics and the consequences of the systematic processing of algorithmic misinformation, HS theory can be applied in the context of misinformation to extend the ongoing information processing model.

Along these lines, the concept of diagnosticity has recently been proposed to bolster the HS model of information processing in the context of misinformation. Diagnosticity is defined as the extent to which people interpret a piece of information as useful and relevant according to their judgment and decision-making (Ahluwalia et al., 2001). Research over the years has found diagnosticity to be the underlying driving factor behind the attitudes and behavioral decisions of users (Kwon et al., 2020; Shin, 2024). In algorithmic contexts, information that is perceived to be highly diagnostic is given more weight in users' judgments and evaluations (Shin, 2023); thus, diagnosticity can be one factor that influences how users systematically and/or heuristically process messages when using GenAI. More specifically, in GenAI, diagnosticity can determine the degree to which users believe that information is true when evaluating an AI-generated content. Upon realizing the diagnosticity of a system, users may trust it and begin to systematically assess misinformation by measuring its accuracy and credibility (Chen and Cheng, 2020; Vu and Chen, 2023). The indicators for diagnosticity could be specific to tasks and goals, and the concept of diagnosticity could be operationalized according to tasks and goals (Shin, 2024). Regarding misinformation from AI-powered media, the indicators of diagnosticity are uniquely specific to users' understanding of the accuracy and credibility of information (i.e. the correctness and reliability of the suggested information). Based

on these insights from previous research, Shin (2023) proposed a conceptual model to explain the heuristic processing of users and how the perceived diagnosticity of misinformation may influence the intentions of users who are exposed to misinformation through algorithms. Numerous studies have expanded on the theoretical foundations of misinformation using information processing theory and diagnostic evaluation (e.g. Chen and Cheng, 2020).

Research questions: a conceptual characterization of generative misinformation

With the popularity of GenAI and the prevalence of misinformation spread through generative mechanisms, determining how people construe and process misinformation when responding to GenAI-produced content is an important area of inquiry (Shin, 2023). Previous research on countering misinformation has primarily addressed the degree to which reason and reflection deter or facilitate the development of correct assessments (Borukhson et al., 2022). For example, an information-oriented approach views misinformation according to an information deficit principle that focuses on information-related factors, a lack of user access to correct information, or the provision of more information to users (Ecker et al., 2022). This approach has produced useful results, but it potentially underplays the social, cognitive, and psychological dimensions of heuristic formation and systematic evaluations. To illustrate, in some cases, people may deny the pandemics of respiratory viruses despite the proven scientific proof. This rejection of publicly accepted information is not the result of a lack of information; rather, it is driven by psychological and social factors, such as conspiracy theories, amplified bias, and selective cognition, which are influenced by individual or personal values instead of factual information (Islam et al., 2020). Psychological processes are influenced by motivation, cognition, understanding, perception, and beliefs, which can be grouped into heuristic and systematic processes. Understanding the relationship between different processes is essential to grasping the context of health decision-making. While extant research suggests that diagnosticity is the underlying driving factor in the attitudes and behavioral decisions of users (Kwon et al., 2020; Yi et al., 2017), the ways in which diagnosticity is formed, construed and practiced by users in the HS process should be examined. Thus, the following research questions (RQs) guide this study:

RQ1. What are the cognitive mechanisms of misinformation's effects on users' use of GenAI?

RQ2. How do users detect misinformation within GenAI?

RQ3. What are the dynamic mechanisms through which misinformation is evaluated and disseminated in the context of GenAI?

RQ4. How do users construe a sense of diagnosticity in algorithmically generated information, and what is the role of diagnosticity in the HS process?

Literature review

Underlying cognitive processes of misinformation evaluation

The proposed model is rooted in information-processing theory and explores the HS processing of misinformation in AI systems. Our hypotheses help investigate the sense-making of users when they are presented with misinformation generated by algorithms and the effects of such sense-making on their sharing behavior and misinformation discernment.

Heuristic processing

In response to the growing public attention directed toward ChatGPT, people's awareness of the potential misuse of GenAI has increased (Van Dis et al., 2023). Awareness of people's individual understandings of AI has also increased, as the specifics of GenAI or internal algorithms are not easily comprehensible to the public due to their black-box nature (Ali et al., 2022). Under these AI-related circumstances, users face high levels of uncertainty and rely on their own heuristics (Tversky and Kahneman, 1974). In normal contexts, users gather and use relevant information from available sources to make optimal choices. When an information decision (whether a given piece of information is valid) is generated in AI contexts, which are characterized by opaque processes and possible biases, users may change their evaluations, effectively adjusting the way in which they evaluate and discern misinformation. The heuristic assessment of misinformation involves evaluating the intrinsic nature of algorithmic attributes, such as fairness, accountability and transparency (FAccT). Growing attention has been given to how to include the principles of FAccT in AI (Diakopoulos and Koliska, 2017; Shin, 2023; Starke et al., 2023). Fairness in AI ensures that AI systems do not continue or amplify existing biases and that they treat all individuals fairly and without discrimination. It also involves ensuring that AI systems do not propagate or amplify any new biases. Accountability refers to the process of ensuring that those responsible for the actions and decisions made by AI systems are held accountable and that processes are in place to resolve any negative effects or harm that may result from their actions. Transparency principles ensure that AI systems are transparent in their operations and decision-making processes and that they can be interpreted and understood by humans. Specifically, this refers to the fact that AI systems should be able to communicate their reasoning to humans.

Improving FAccT and its performance has been an important concern for people and the public (Diakopoulos and Koliska, 2017). Individual users have begun to use the FAccT framework to evaluate AI systems (Gran et al., 2021). As users are unable to evaluate AI systematically, they rely on heuristic cues, such as fairness, transparency, and trustworthiness, to process large amounts of generated information. For example, as Sundar et al. (2007) showed that news chatbot users rely on heuristic cues to process recommended news feeds, GenAI users apply heuristics before systematic evaluations to process the information generated by the AI. Users want to verify the FAccT of the systems that spread misinformation (Shin, 2023). Examining the ethical/normative factors

(i.e. fairness, bias, equitability, and auditability) of the methods developed to counter the dissemination of misinformation is critical.

Fairness heuristics. Because users' cognitive ability to process the increasing amount of misinformation in generative content is limited, they normally apply fairness heuristics when interacting with GenAI. For example, people use their own heuristics to judge possible algorithmic biases, such as ethnicity and gender biases, from GenAI (Mhasawade et al., 2021). As people rely more on GenAI for important decisions and relevant information, biases in AI will become increasingly concerning and may potentially cause harm. In reality, users have a limited capacity to judge algorithmic biases, as they can neither evaluate message validity on their own nor systematically configure and correct biases by tracing them in data and internal algorithms. Shin (2023) showed that the algorithmic fairness of AI hinges upon the perceptual impartiality and equitability of its users. Related research has clarified the causal relationship between algorithmic fairness and user attitudes, particularly as it relates to user efficacy and confidence in evaluation (Starke et al., 2023). Because the relationship between the fairness of algorithms as perceived by users and diagnosticity has been confirmed in prior research, the following connection between perceived algorithmic fairness and the diagnosticity of AI-generated content can be proposed:

H1. Users' fairness heuristics positively affect perceived diagnosticity during their interactions with GenAI.

Accountability heuristics. Algorithmic accountability is the idea of whether information is shared under substantive and procedural standards that hold entities accountable if they fall short of these standards (Diakopoulos and Koliska, 2017). Accountability is particularly important in monitoring the spread of misinformation, as the repercussions of the spread, amplification, and diffusion of misinformation are immense. People may believe that GenAI should be held accountable for disseminating misinformation and must provide further clarification when requested by the public. Related research shows that users develop positive confidence with clearly defined responsibilities and liabilities when they correctly judge incorrect information as misinformation (Barnoy and Reich, 2022). Based on previous findings, examining how users construe algorithmic accountability in relation to misinformation and how their perceptions influence their diagnosticity of misinformation in GenAI is relevant. Therefore, the following hypothesis is worth considering:

H2. Users' accountability heuristics positively influence their diagnosticity of misinformation in GenAI.

Transparency heuristics. Transparency in AI requires that both the inputs and outputs of algorithms be visible and comprehensible to users or the public (Shin, 2023). For example, the transparency of data acquisition cues can be heuristic when evaluating data protection in AI. Because end users generally do not have the ability to examine the

transparency of algorithms, users base their judgments on easily processed heuristics when the related heuristic cues are available (Tversky and Kahneman, 1974). For instance, transparent statements showing the processes of generative mechanisms in websites or AI systems may provide users with certain heuristics in which they can quickly form their judgments on transparency without engaging in extensive technical evaluations of the systems or platforms. As users are unable to process each piece of information systematically or each component of the AI system technically, users depend on their own heuristics according to the available cues (Molina and Sundar, 2023). When users have a sense of transparency in systems, they frequently deem the information from the systems trustworthy and reliable (Gran et al., 2021). Research has confirmed that transparency enhances user confidence in an algorithm and plays a key role in users' usage and adoption of the information generated from algorithms (Peifer and Meisinger, 2021). When transparent and visible cues are available, people are more likely to view the provided information as diagnosable (Diakopoulos and Koliska, 2017). Thus, it can be proposed that transparent AI systems can afford users a sense of assurance and prepare them to diagnose misinformation. Thus, the following hypothesis is worth considering:

H3. Users' transparency heuristics positively influence the diagnosticity of GenAI.

Diagnosticity: how users form their efficacy. When arriving at a diagnostic judgment about the misinformation and systems of the corresponding heuristics in cognition, users evaluate both the content of the information and the systematic cues it contains (Niu et al., 2021). Having diagnostic information is important for AI to provide trustworthy services. Diagnosticity is defined as the extent to which people interpret a piece of information to be truthful and relevant according to their judgment and decision-making (Ahluwalia et al., 2001; Chen and Cheng, 2020). In AI contexts, diagnosticity refers to a user's perceived ability to understand the features and attributes of AI services in their decision-making (Shin, 2023). Estimating the diagnosticity of information from AI must be based on inference and personal speculation. Because information that is perceived to be diagnostic is given relatively more weight in users' judgments and evaluations (Yi et al., 2017), diagnosticity influences how users systematically process messages (Niu et al., 2021). Upon realizing a system's diagnosticity, users can grow to trust it and begin to systematically assess misinformation by measuring its accuracy and credibility (Chen and Cheng, 2020; Stecula et al., 2020). The indicators of diagnosticity are uniquely specific to users' ability to identify misinformation based on their understanding of the information's accuracy and credibility (i.e. the correctness and reliability of the suggested information; Vu and Chen, 2023; Zrnek et al., 2022). It can be hypothesized that diagnosticity functions as a conduit between heuristic and systematic evaluations. Therefore, the following hypotheses are worth testing:

H4. The diagnosticity perceived by users positively affects the credibility of information.

H5. The diagnosticity perceived by users positively affects the accuracy of information.

Systematic processing

Systematic processing involves efforts to understand available information through careful attention, complex analytics, deep thinking, and intensive reasoning (Todorov et al., 2002). It not only encompasses active comprehension and evaluation of a message's arguments but also includes verifying the information's validity leading to the message's conclusion. Accordingly, this process results in an informed evaluative judgment and decision (Shin and Park, 2019), as well as greater confidence regarding the adoption decision (Shin, 2020). Within AI systems, users employing systematic processing critically assess information by evaluating its accuracy and credibility (Ali et al., 2022). This involves a performative evaluation of personalization and accuracy. Users rationally consider the trade-offs associated with FAccT against utilitarian factors such as the level of personalization, the degree of accuracy in personalized news. Although systematic processing confers more confidence on users, it demands significantly more effort and resources, including making judgments based on accurate information, evaluating information using explanatory cues, or assessing credibility using credible source (Peng and Wang, 2023).

The accuracy of generated information. The perceived accuracy is a user's perception of the amount of misinformation present in the information they encounter (Pennycook and Rand, 2022). It reflects the user's judgment of the truthfulness of the present information. In the context of GenAI application, accuracy is related to correctness of generated content and the veracity of the recommended content. As for the measurement of the perceived accuracy of misinformation, relevant studies have proposed various terms to describe it, including misinformation awareness, which is a person's awareness that the received information is false (Schuetz et al., 2021). Empirical research across various algorithmic services has established accuracy as a determining factor influencing attitude and behavioral intention (Amoozadeh et al., 2023; Pennycook and Rand, 2022; Shin, 2023). Amoozadeh et al. (2023) found a significant correlation between trust in GenAI and its impact on students' motivation and confidence in programming. Their research suggests that the perceived accuracy and reliability of GenAI tools significantly affect users' trust, which in turn, could influence their intention to discern misinformation. Accordingly, the following relationship is proposed:

H6. Users' perception of accuracy positively influences users' intention to discern misinformation (the more users see the importance of accuracy, the more likely they evaluate misinformation consciously).

The credibility of generated information. As a strong predictor of the influence of information, credibility can generally be understood to refer to believability and trustworthiness (Wu et al., 2022). In the communication literature, credibility is measured according to

different perspectives, such as source credibility, message credibility, and media credibility (Barnoy and Reich, 2022). Although credibility has historically been studied more extensively from the source perspective, news consumers in the social media environment have placed greater attention on the content of a message than on its source (Cappella, 2006; Peifer and Meisinger, 2021). In GenAI, source credibility and content credibility are equally important, as content is generated by AI (Ecker et al., 2022). The more credible a piece of misinformation's content is perceived to be, the more useful and relevant (i.e. diagnostic) users perceive it to be. Thus, the following hypothesis is proposed:

H7. Users' perceptions of the credibility of news misinformation influence their intentions to discern misinformation.

Explanatory cues as interventions: the moderating effect of explainability

Explainability has become a critical component of generative AI, which encompasses technologies like machine learning and AI that provide human-comprehensible rationales for their outputs (Rai, 2020; Shin, 2023). This concept is crucial as AI systems and algorithms, often perceived as complex "black boxes," require high levels of specialized expertise and knowledge to understand their decision-making processes or performance of AI systems (Rai, 2020). ChatGPT, as it currently stands, is an example of non-explainable AI: While users might have a general understanding of how the answers were generated based on the prompt/input, the specifics of how and why it yields certain answers remain opaque (Shin, 2023).

As security, safety, or explainability are becoming critical requirements in AI, it becomes imperative for AI systems like ChatGPT to incorporate these functions and features. From the user side, responsible use of AI systems, coupled with an awareness of its limitations, is also important. When taking into account the explainability, ChatGPT could evolve into a knowledge reference model that aids in the extension of cognitive models by generating potentially meaningful additional services. To do this, ChatGPT should be equipped with explanatory cues and linking information to verified sources in a traceable manner.

While similar in concept, transparency and explainability are different in definition and usage. Transparency in AI systems enables users to understand how a GenAI system is designed, trained, performed and deployed in a relevant domain so that users can make more informed decisions. It refers to the capacity to provide relevant information and clarify what information is generated and why. In contrast, explainability pertains to the AI system's capacity to enable users of an AI system to understand how certain information is generated, through giving easy-to-understand information, including explanatory cues or supplementary details.

Diagnosticity and explainability, while serving different purposes, are both integral to user trust and comprehension, especially concerning misinformation. Diagnosticity pertains to users' capability to evaluate misinformation by considering particular aspects of information. In contrast, explainability refers to a model's or system's ability to describe

its generation processes. The two are fundamental in building trust, credibility and understanding between an AI agent and its user, especially in the context of misinformation (Rai, 2020).

The significant roles of explainability and diagnosticity in user perception have been demonstrated in recent studies (Schrills and Franke, 2023; Siering, 2022). Schrills and Franke's (2023) research regarding automated insulin delivery systems underscores the significance of explainability in AI systems, asserting the necessity of traceability in automated decisions for effective user interaction. They highlight the concept of subjective information-processing awareness (SIPA) as a crucial component in understanding users' experiences with explainable AI. Their research reveals that different levels of information disclosure affect SIPA and, consequently, user performance and that SIPA is strongly correlated with trust with explanations, which underscores the importance of explainability in building user trust. The insights from Schrills and Franke's study provide a meaningful context for understanding the role of explainability in GenAI systems like ChatGPT.

Given the significant effects of explainability and diagnosticity explored in previous studies, we propose examining the moderating role of explainability in the evaluation of misinformation in ChatGPT.

H_{m1} . The effect of users' perceived diagnosticity of misinformation on their perception of its accuracy is moderated by the explainability of the information generated by GenAI.

H_{m2} . The effect of users' perceived diagnosticity of misinformation on their perception of its credibility is moderated by the explainability of the information generated by GenAI.

Based on the constructs and hypotheses presented in this study, we propose a conceptual model that depicts how users process misinformation related to the harmful effects of AI-generated content, its antecedents and its consequences (see Figure 1).

Methods

We used experimental surveys with the Wizard of Oz (WOZ) method to empirically investigate people's cognitive processing of misinformation. The *WOZ* is a user-research *method* that involves interaction with a mock interface controlled by a human. A pilot study was conducted with 30 participants prior to the main experiment in order to ensure the reliability of the WOZ method. When finished with the pilot test, six were selected again out of the 30 participants for the pretest. This two-tier process made sure to test the consistency of the method. Six research participants completed a pretest questionnaire to check its reliability and validity. Recommendations and preliminary results from this pretest were incorporated in designing the interface and the questionnaire. A finalized WOZ experiment was set up in which participants interacted with the simulated ChatGPT. The participants believed that the interaction was operated and controlled by GenAI but operated by human agents (research team). The research team sat behind the wizards and acted as if GenAI. When respondents inquired about certain questions, the research team

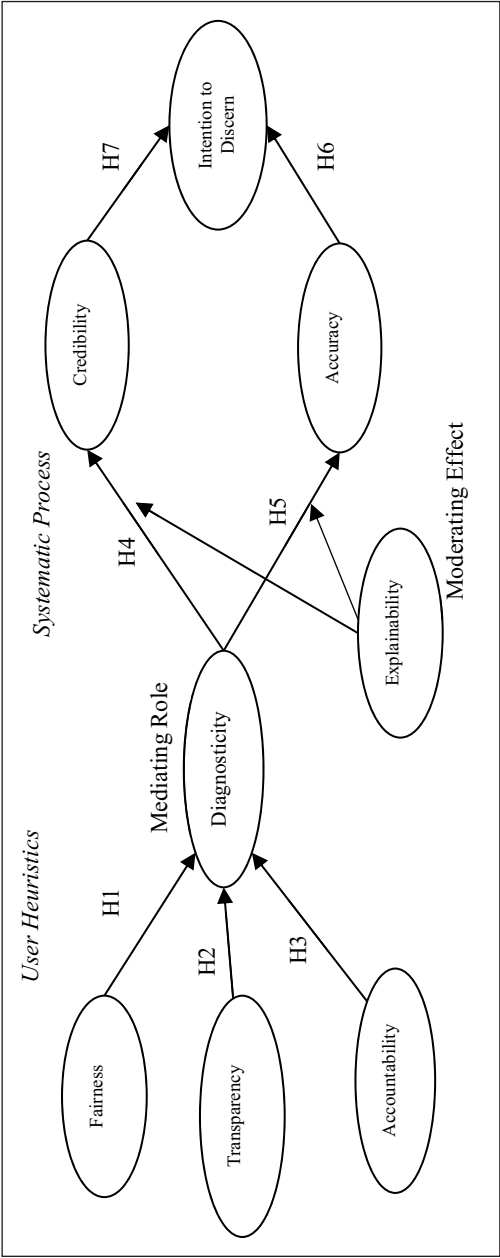


Figure 1. Conceptual model.

obtained the questions by asking ChatGPT on their ends and returning the response to the respondents (by typing the answers manually). When the research team responded with misinformation, the research team made up the answers intentionally containing misinformation or incorrect information. The interface was designed to look like a real ChatGPT, and respondents believed it was ChatGPT or a variation of GenAI.

Collection of data and samples

The WOZ method was used to examine users' misinformation processing. The WOZ interface was created for users to feel like they were communicating with a GenAI application—ChatGPT (Figure 2). We simulated interaction between users and GenAI for the purpose of data collection. Participants used the WOZ GenAI as if they were using ChatGPT. As the participants chatted and questioned the GenAI, human wizards responded with appropriate questions that resembled the AI's responses.

After receiving approval from the Institutional Review Board of Zayed University, the study was administered to 302 respondents from January to April 2023 in the United Arab Emirates. Respondents were offered either course credits or monetary incentives for participation. The data were collected through a two-tier process. First, from January to February 2023, initial respondents (151) were recruited from college students in the UAE (Dubai, Abu Dhabi, Al Ain, and Sharjah). These respondents participated in the experiment from the courses in digital journalism and new media. They received course credits for completing the study. The experimental facility was set up at the computing center at Zayed University for those respondents who preferred offline experiments. Participants could visit the physical lab with the computer set up to take the experiment. The faculty and staff participated in the experiment. The median time for completing the study was 22 minutes ($M=20.19$, $SD=7.92$).

The second round of data collection was done from March to April 2023 for the public online. In this round, respondents were recruited from Qualtrics panels, and panel members were invited to participate in the survey through Qualtrics, which offered monetary incentives for completing the experiment. Participant samples were recruited from the country to ensure a similar demographic profile to the first round of data. Upon obtaining informed consent, the participants were asked to visit the setup link, where they could see the interface of the WOZ GenAI. The median time for completing the online study was 25 minutes ($M=21.27$, $SD=6.87$).

The interface, process, and stimuli of the two modes of experiment were the same. After receiving brief instructions, the participants were given a set of predefined tasks and were asked to fulfill them by interacting with human wizards. They were provided with a predetermined list of news items on current health and health/scientific campaigns with related misinformation. Participants could stop and ask clarifying questions or leave anytime they wanted. Respondents recruited for the study ($n=302$) were randomly assigned into two conditions, with no statistically significant differences among the conditions regarding gender, education, prior knowledge about algorithms or media consumption of news: control group without stimuli ($n=151$) and experimental group with explanatory cues ($n=151$). We equalized the number of participants in the two conditions to compare the effects of control and experimental conditions. They were then asked to interact with the GenAI through

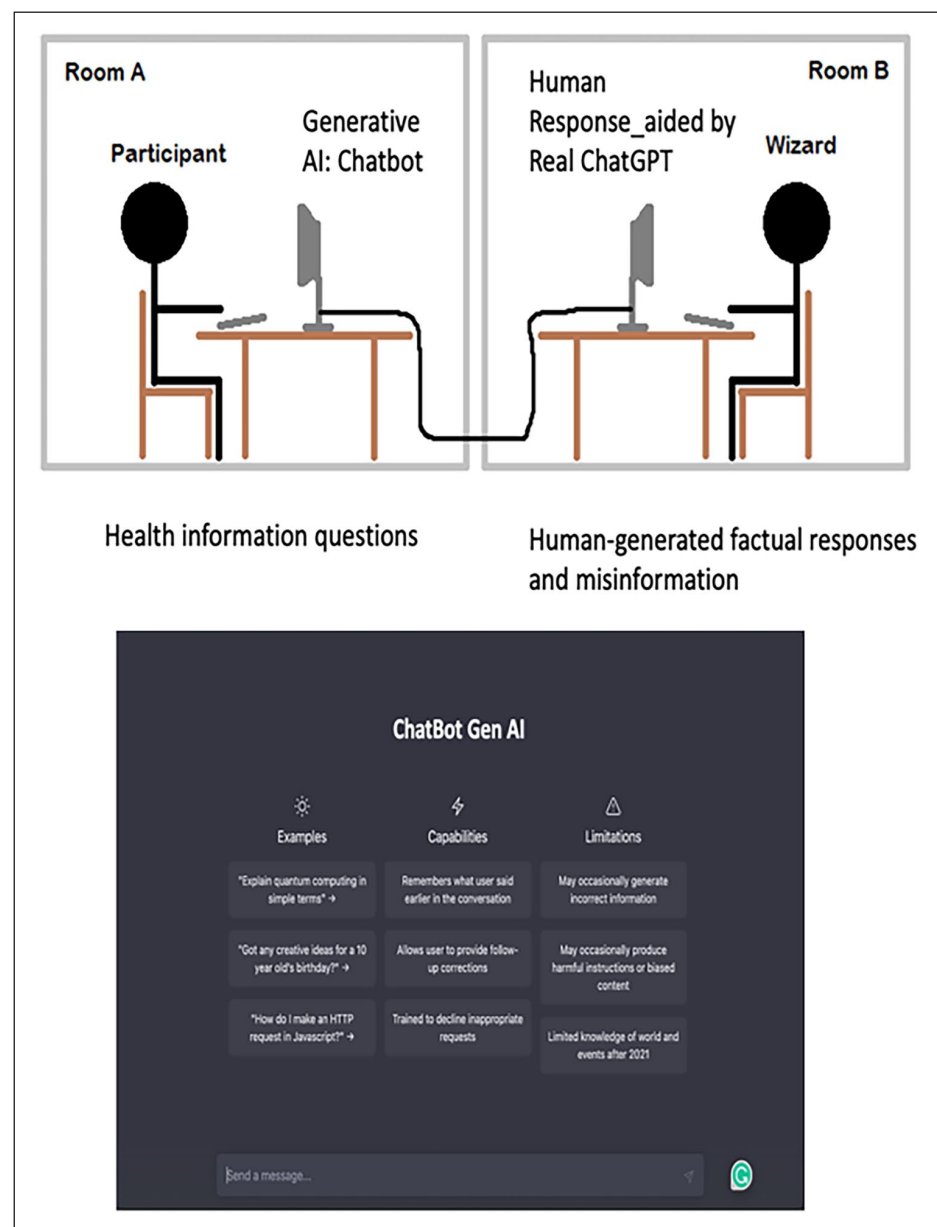


Figure 2. Experimental setup for the GenAI WOZ method.
The interface of the experimental GenAI.

the interface and read current health-related news (health information currently being discussed and proliferated online) the wizards had generated for approximately 20–30 minutes. Examples of the covered health topics included COVID-19,

infectious viruses, respiratory symptoms, vaccines, mental health awareness, and climate change. The human wizards played the same role as GenAI, as they used ChatGPT to answer the participants' questions and, at times, responded with misinformation, fake news, incorrect recommendations, or misinformation related to the questions. The ratio of true information to misinformation was equal to 10:1.

The WOZ GenAI provided the participants with contemporary news containing both true information and misinformation on contemporary health issues and medical campaigns. The human wizards were instructed to provide misinformation during interactions with the respondents. The WOZ method was designed to test how users perceive and process the misinformation suggested by GenAI. After interacting with the GenAI (chat, inquire, search, and respond), the participants completed a survey (Table 1).

Data analyses

The proposed model was tested through a two-stage method used by Islam et al. (2020) to predict misinformation behavior. During the first stage, a structural equation modeling (SEM) method was employed to confirm the validity and reliability of the constructs and to test the structural paths between them. Next, a neural network (NN)-based method was applied because SEM cannot examine the nonlinear paths between constructs, which can be important in AI contexts. The NN analysis was designed to validate the SEM results and to prioritize predictors based on their relative importance in affecting HS processes. The model tested in this study was the outcome of the experiment. The data from the experiment were subsequently reused for the NN method.

Scales and measurements

The scales used in this study comprised 21 measurements with a 7-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). All measurements were based on the literature on health communications and research on misinformation (Appendix 1). The measurements were administered using combinations of previously developed metrics and modified metrics from existing studies. Such combinations were necessary because certain measurements in our analysis needed to be changed to reflect the distinctive features of AI algorithms and services.

These measurements were then compared with measurements from the literature on AI, communications, and cognitive science. The FAccT, explainability, and diagnosticity measurements were performed based on the works of Shin (2023), Rai (2020) and Chen and Cheng (2020), respectively. Some measurements required changes to reflect new traits in the misinformation. Explanatory stimuli were presented as cues in the misinformation, popping up as phrases when the subjects clicked on the recommended news. These cues provided users with relevant explanations, such as "because you searched it before" or "most similar [to] viewed news," on why specific health information was recommended. The explanatory cues allowed participants to understand, interpret, and experiment with the misinformation.

Table 1. Descriptive statistics (*n* = 302).

Characteristics	Sample
Age (mean/SD/median)	25.30/1.10/29
Gender (female rate) (%)	50.98
College educated (%)	33.18
AI platform experience	2.8 years

Table 2. Discriminant validity.

Construct	1	2	3	4	5	6	7	8
Fairness	0.71							
Accountability	0.21	0.83						
Transparency	0.33	0.52	0.83					
Accuracy	0.13	0.23	0.52	0.93				
Credibility	0.12	0.11	0.23	0.42	0.72			
Diagnosticity	0.23	0.53	0.13	0.52	0.11	0.82		
Explainability	0.42	0.22	0.53	0.21	0.53	0.13	0.82	
Intention	0.52	0.11	0.22	0.22	0.22	0.53	0.23	0.73

Measurement instrument

The validity and reliability of the measurement items were assessed based on the reliability and convergent validity criteria using Analysis of Moment Structure (AMOS) 18 software. The Cronbach’s alpha of the survey instrument was calculated to test its reliability. The analysis results were verified, and varimax rotation was performed on the 30 original items (seven items were eliminated due to low loading). According to Hair et al. (1995), an item has high loading if its loading coefficient is above 0.6. In this study, each item’s loading value was above the recommended level of 0.8 (Cronbach, 1989). Confirmatory factor analysis was performed to test the validity of the instruments; furthermore, the composite reliability, estimated percentage of variance extracted by each construct, and discriminant and convergent validity were examined. All the constructs in the model met the required reliability and convergent validity values (Tables 2 and 3). Regarding discriminant validity, the average variance extracted (AVE) values and the correlations of the factors were examined. The shared variance between the constructs was lower than the AVE from the individual constructs, confirming the presence of discriminant validity. Correlation analysis was performed, and Pearson’s *R* (correlation coefficient) showed an acceptable level of correlation among the variables. In conclusion, the measurement model demonstrated adequate reliability and the presence of convergent and discriminant validity.

Fit indices. The structural model was tested using the AMOS procedure and maximum likelihood-based SEM software. The results indicated the presence of no misfits (Table 4). All values were within acceptable ranges, indicating good overall model performance.

Table 3. Reliability checks for constructs.

Construct	Initial items	Final items	Mean (SD)	Cronbach's alpha
Fairness	4	3	4.17 (1.011) 3.84 (1.055) 4.11 (1.047)	0.795
Accountability	4	3	4.17 (1.011) 3.84 (1.055) 4.11 (1.047)	0.619
Transparency	5	3	3.39 (1.054) 3.98 (1.128) 3.30 (1.282)	0.875
Accuracy	4	3	3.74 (1.346) 3.87 (1.385) 3.40 (1.411)	0.812
Credibility	4	3	4.37 (1.182) 4.30 (1.141) 4.63 (1.202)	0.842
Diagnosticity	4	3	4.37 (1.182) 4.30 (1.141) 4.63 (1.202)	0.853
Explainability	3	3	4.34 (1.210) 4.44 (1.295) 4.05 (0.984)	0.856
Intention	3	3	4.42 (1.121) 4.10 (1.264) 4.18 (1.268)	0.900

The Comparative Fit Index (CFI) and Tucker Lewis Index (TLI) values appeared low relative to the other guidelines (.95 and over). Increasing the number of indicators (p) led to a decline in the average sample estimates of CFI and TLI, indicating that the model fit worsened. However, the average root mean square error of approximation (RMSEA) tended to decrease (indicating an improved fit) as more indicators were added to the correctly specified models. Thus, the model fit indices suggest a good fit.

Structural model testing

The SEM results validated all the relationships in the hypotheses (Table 5). The proposed path coefficients were statistically significant ($p < .001$ or $p < .0001$). Moreover, the perceived diagnosticity was significantly affected by heuristic evaluations and simultaneously influenced the systematic processing of misinformation. The heuristic factors accounted for 59.1% of the variance in diagnosticity. The intention to share information was significantly influenced by the systematic processing of misinformation, with systematic factors explaining 32% of the variance in the intention to share information. The model explained a significant percentage of the variance in each factor. Furthermore, the

Table 4. Model fit indices.

Fit statistics	Structural model	Suggested value (Joreskog and Sorbom, 1996)
χ^2 (df)	1710.5/244=7.01	<5
p value	0.000	<0.05
RMSEA	0.023	0.05 < x < 0.10
CFI	0.910	>0.90
NFI	0.978	>0.90
GFI	0.920	>0.90
IFI	0.802	>0.80
TLI	0.901	>0.85
RFI	0.803	>0.80
AIC	1870	
Hoelter's critical N	66(.05)/70(.01)	Higher values signify a satisfactory fit
ECVI	4.688	Smaller values denote a better fit

Table 5. SEM results.

Paths	Estimate	Standard error	Critical ratio	p value
H1: Fairness→Diagnosticity	0.610	0.129	4.739	***
H2: Accountability→Diagnosticity	0.512	0.043	11.840	***
H3: Transparency→Diagnosticity	0.457	0.047	9.791	***
H4: Diagnosticity→Accuracy	0.642	0.067	9.517	***
H5: Diagnosticity→Credibility	0.649	0.076	8.499	***
H6: Accuracy→Intention	0.396	0.063	6.292	***
H7: Credibility→Intention	0.221	0.087	2.525	.012*

*1.96: 95% (0.05), **2.58: 99% (0.01), & ***3.29: 99.9% (0.001).

significant paths indicated a causal conceptual link between the heuristic and systematic processes through diagnosticity (Table 5).

Neural network analysis

Table 6 lists the average importance and average normalized importance of each input factor in predicting diagnosticity and sharing intention. Transparency is the most critical predictor, followed by fairness and accountability, in predicting diagnosticity (heuristic process). Explainability is the most determining factor in forming a systematic process (sharing intention), followed by credibility and accuracy. The results are largely consistent with the SEM results.

The impact of explanatory heuristics on systematic evaluation

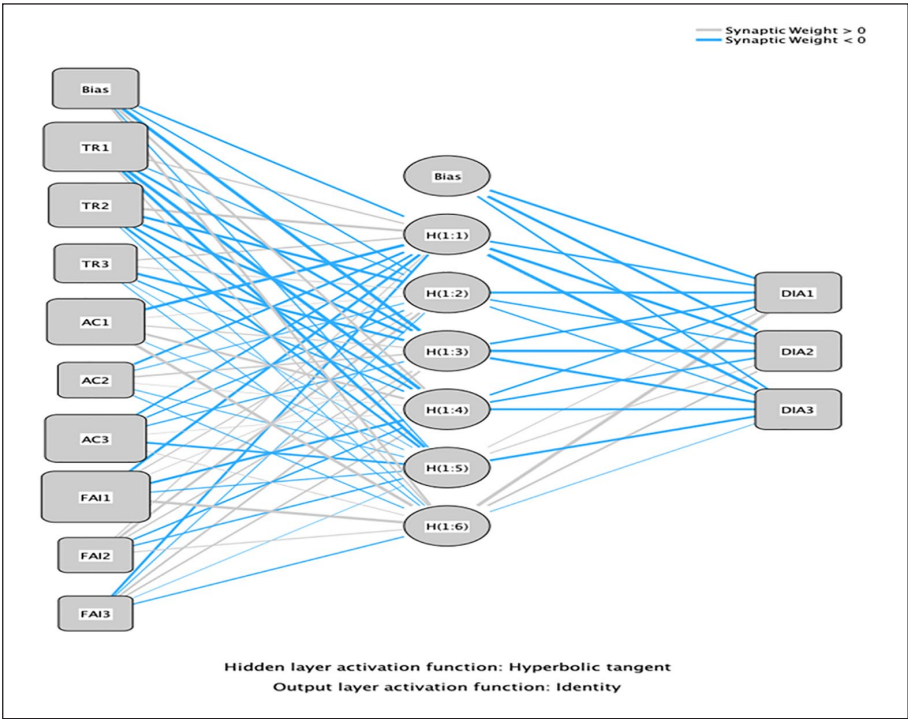
We examined the moderating effects of explanatory cues on the HS processing of diagnosticity in the model. The explanatory cues in the GenAI provided users with

explanations of why and how the recommended information were generated. They also helped users systematically assess misinformation. The investigation of the moderating roles of explainability and diagnosticity on accuracy and credibility was worthwhile. We conducted a multigroup analysis to evaluate the structural paths from the model to the control group and experimental group (Table 7).

In the experiment, explanatory cues were provided along with generated information during the participants' chats with the human wizard of the WOZ GenAI, whose interface was designed to resemble that of ChatGPT. Such explanatory cues included

Table 6. Neural network-based approach for predicting heuristic process.

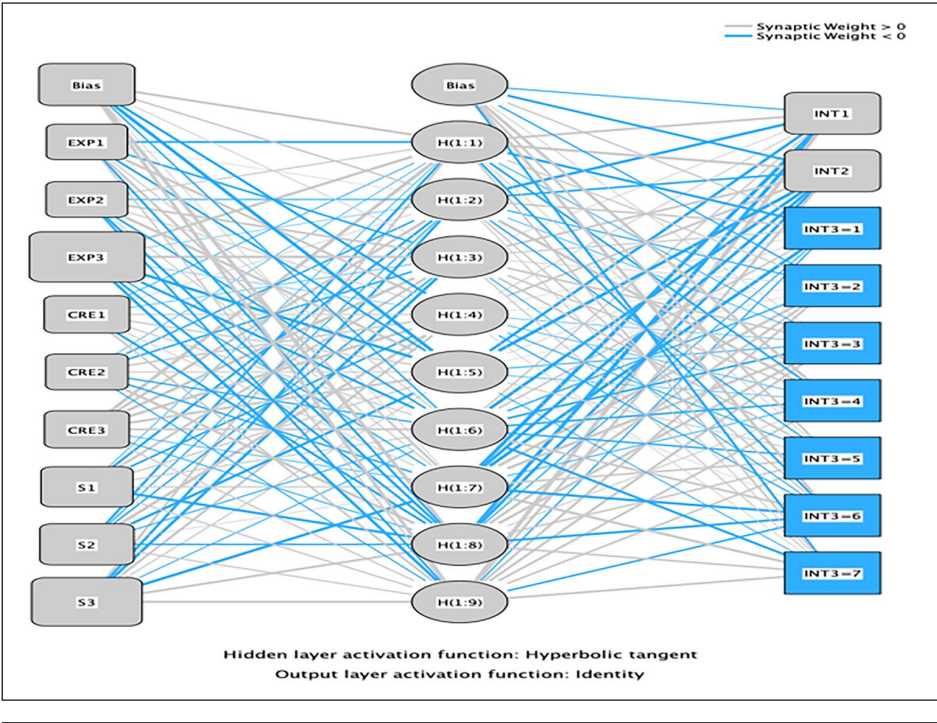
Predictor heuristic	Heuristic items	Normalized importance	Rank
Transparency	TR1	90.4	1
	TR2	68.2	
	TR3	42.8	
Fairness	FAI1	98.0	2
	FAI2	25.0	
	FAI3	24.4	
Accountability	AC1	76.3	3
	AC2	27.5	
	AC3	84.2	



(Continued)

Table 6. (Continued)

Predictor heuristic	Heuristic items	Normalized importance	Rank
Explainability	EXP1	11.5	1
	EXP2	12.7	
	EXP3	99.0	
Accuracy	ACC1	39.7	2
	ACC2	44.9	
	ACC3	88.0	
Credibility	CRE1	22.1	3
	CRE2	14.2	
	CRE3	20.9	



fact-checked information, information sources, accuracy nudges, and brief explanations of how and why certain health information was generated. The experiment’s participants were paid, and the participation time was approximately 20 minutes ($M=21.28$, $SD=6.42$).

The two groups showed equally good fits with the data, and distinct patterns were observed that supported the hypotheses. The results showed notable distinctions in item composition and path formation, providing insights into the effects of explainability on misinformation (Table 7). The effects of diagnosticity on credibility

Table 7. Moderating effects of explainability.

Path	Control group					Explanatory group				
	β	SE	CR	p	Result	β	SE	CR	p	Result
Diagnosticity–credibility	.066	.041	1.592	.111	No	.684***	.050	13.727	.000	Yes
Diagnosticity–accuracy	.037	.050	.750	.454	No	.106***	.032	3.301	.000	Yes
Accuracy–intention	.166*	.113	1.434	.005	Yes	.328***	.021	1.734	.000	Yes
Credibility–intention	.337**	.054	.342	.01	Yes	.421***	.081	3.492	.000	Yes

* $p < .05$, ** $p < .01$, & *** $p < .001$.
 β : Unstandardized coefficient.

and accuracy were more significant in experimental group than in control group. In comparing the two models, experimental group showed higher values for systematic evaluations, and control group showed higher values for heuristic evaluations. Therefore, we can infer that explanatory cues played a facilitating role in the systematic evaluations and heuristic processes and that their effects were greater on the systematic evaluations than on the heuristic processes. The results of the squared multiple correlations also support the underlying role of explainability (Table 7). The R^2 values of accuracy and credibility in control group were greater than those in experimental group. Moreover, the R^2 value of intention in experimental group was distinctly higher than that in control group.

Discussion

AI-generated misinformation and human-generated heuristics

As proposed by Tversky and Kahneman (1974), using suitable heuristics can help users engage in various cognitive processes and lead to correct diagnostics of misinformation generated by AI. The results of the current study showed that the users’ HS processing significantly influenced their attitudes and behaviors. The users’ misinformation processing was heuristic-based and systematically oriented, and the GenAI users processed generated information with available heuristics and systematic cues to arrive at a diagnostic judgment. We found that the more users felt they could detect and evaluate misinformation, the more likely they were to determine the accuracy and credibility of information. Moreover, the more likely users felt that the information was systematically reliable (credible or accurate), the more likely they were to accept it. Heuristic knowledge of the misinformation significantly influenced users’ construed diagnosticity and the subsequent credibility of that information. Furthermore, the explanatory cues of the information moderated the effects of diagnosticity on the perceived accuracy and credibility of that information. HS processing was mediated by perceived diagnosticity and moderated by the extent of the explainability embedded in the AI system. Together, explainability and diagnosticity were the key user heuristics through which misinformation influenced users’ discernibility and intention behaviors.

Theoretical implications: algorithmic effects on misinformation

This study extends HS theory by applying it to the context of misinformation generated by GenAI and examining the antecedents and consequences of diagnosticity. The results offer relevant theoretical implications for algorithmic effects on health misinformation (Peng and Wang, 2023). First, although previous studies have researched the HS model, this study elaborated on the duality of the information-processing mode when users consider generated misinformation. The study characterized the dual-process misinformation as follows: a circumstance where users process misinformation heuristically, and such heuristics invoke a sense of diagnosticity, facilitating systematic evaluation. It was also determined that heuristics call upon prior knowledge and lower cognitive load when users interact with GenAIs. Diagnosticity is known to serve as a motivator or facilitator for heuristic processes and to link them to systematic processes. In other words, users' existing prior knowledge plays a key role in bringing heuristics into the evaluation process (Todorov et al., 2002), and diagnosticity serves as a catalyst for the further evaluation of GenAI (Yi et al., 2017) in terms of systematic evaluation.

One of this study's significant theoretical contributions is in its identification of a new set of heuristics that emerged during the interactions with GenAI. Specific heuristics occur in misinformation judgment contexts. For instance, FAcT heuristics occur because users want to know the normative values of AI, and diagnostic heuristics occur as a motivation to further investigate AI systematically. Thus far, there remains a gap between heuristic and systematic processing. By identifying the role of diagnosticity, this study linked heuristic processing to systematic processing in the GenAI context. The dual- or two-tier process, along with the mediating role of diagnosticity, can fill the gap in research on user behaviors related to misinformation that previous studies have attempted to elucidate (e.g. Ahmed and Gil-Lopez, 2022; Cho et al., 2020; Hwang and Jeong, 2023; Kim et al., 2020; Melchior and Oliveira, 2022; Molina and Sundar, 2023; Tully et al., 2020). Research has also failed to clarify the nature of user engagement and its matrix and process as previous research has primarily examined the unidimensional or monolithic effects of the psychological impact of misinformation on user behavior (e.g. Lazer et al., 2018). We have extended the current understanding of the relationship between user health behavior and misinformation by explicating HS processing by users during evaluations of misinformation. We adopted an information-processing approach that considers the cognitive and psychological development of users and how HS processing during evaluations influences the discernment of misinformation. Our approach revealed relevant factors about the cognitive–psychological processes and helped enrich the understanding of the relationship between how misinformation forges the attitudes of users and how their attitudes toward the information and source of algorithm-generated misinformation influence the emergence of misinformation; this coevolving relation was mediated and facilitated by diagnosticity. Whereas previous studies have merely conceptualized the existence of diagnosticity (e.g. Chen and Cheng, 2020; Yi et al., 2017), we provided the following hypothetical and empirical clarification on the underlying relation of diagnosticity and GenAI features: when a piece of information is generated by algorithms in transparent, fair, and accountable ways, users perceive it with a high level of diagnosticity, which improves their intention to systematically analyze that information.

The mediating role of diagnosticity in relaying the heuristic process to the systematic process is noteworthy. Previous research on HS theory has assumed two separate models of different processes in which disconnected processes have little to do with each other or little to do with interactional relations (e.g. Chaiken et al., 1996; Todorov et al., 2002). In the AI-generated misinformation context, diagnosticity plays a critical role in the following two ways: (1) due to the black-box nature of GenAI generation, users evaluate the attributes intuitively and heuristically, and (2) given the harmful effects of misinformation, users evaluate the generated information holistically and systematically. In this study, the mediating role of diagnosticity revealed that the more confident the GenAI users were about the systems being ethically sound and normatively acceptable, the more likely they were to technologically diagnose the generated information. These results are consistent with earlier findings by Ahmed and Gil-Lopez (2022) and Ali et al. (2022), suggesting that users with lower cognitive heuristics (i.e. a lower understanding of the AI decision-making process or an awareness of fairness/transparency/accountability in AI) are more likely to believe and accept/share misinformation. Our findings are also consistent with ongoing reports of users' understanding of algorithmic functions (Shin, 2023) and related ethical issues (Mhasawade et al., 2021). The results indicated that when algorithm users had a higher level of perceived diagnosticity, they became more confident and capable of analyzing the accuracy and credibility of the information presented through GenAI and, consequently, reacted to the misinformation conscientiously and with a higher cognitive load. The interaction effect this study identified supports the proposition that users who are oriented toward the FAccT heuristic consideration are likely to accept and use explanatory cues in diagnosing the accuracy and credibility of information. This result on the interaction effect highlights a new role of diagnosticity in misinformation processing: diagnosticity in misinformation can result from heuristic processing and from its function as an activator in conducting systematic processing. This argument serves as a relevant theoretical advancement because it clarifies the processing capacity of users through diagnosticity. This role of diagnosticity can be presented as an extension of previous conceptualizations of user awareness (Gran et al., 2021), self-efficacy (Ali et al., 2022), informative fiction (Margolin, 2021), and algorithmic literacy (Shin, 2023). The diagnosticity principle in misinformation may refer to users' heuristics to probe the salience of algorithmic features using available cues and optimal inferences. Users' perceived diagnostics are heuristic-dependent, context-oriented, and can acquire significant diagnostic value if the diagnostic object is extended to include a systematic matrix.

Diagnosticity can be a cognitive trigger for switching from heuristic to systematic processing, functioning as a conduit that channels evaluation efforts from heuristic factors into systematic processes. Normative values give users a sense of diagnostic readiness for handling misinformation, thus leading to an effective, systematic evaluation of misinformation. In GenAI, users construe a sense of diagnosticity based on their understanding of algorithmic heuristics. In other words, the extent of diagnostic ability in AI may reside in users' cognition as much as in salient technological features or performance. Thus, diagnosticity may depend more on how users construe algorithmic heuristics than on actual technological readiness. Diagnosticity is cognitively construed by users' existing cognition, constructed by users' normative values, and practiced according to users' motivations.

This user-centered view of diagnosticity aligns closely with the procedural or generative aspect of misinformation. Although misinformation has been an ongoing problem throughout the COVID-19 pandemic and beyond, relatively few studies have explored the process of misinformation, particularly from users' points of view, with many only studying it as a tool to combat misinformation, such as debunking, prebunking, and inoculation (Hwang and Jeong, 2023; Lewandowsky and Van der Linden, 2021). The scarcity of research on the procedural aspect of misinformation could limit our understanding of how users interact with misinformation (Kim et al., 2020), especially in contexts where GenAI presents health information. Our results confirmed the proposition that diagnosticity is a critical construct (Chen and Cheng, 2020) and further corroborated that HS processing in GenAI concurrently affects the intention of users to process misinformation through dual paths (Pennycook, 2023). Prior work has primarily focused on the unidirectional influence of misinformation from a functional viewpoint, exploring such topics as misinformation traits (Borukhson et al., 2022), with few studies holistically examining the dual processing of misinformation. Our study advances ongoing research by clarifying how heuristic and functional dimensions can increase user diagnosticity and how users evaluate the accuracy and credibility of misinformation.

Practical implications: design of intervention strategies for generated misinformation

Our study contributes to a shared understanding across the AI industry and the broader scientific community of how GenAI can better address integrity issues on its platforms. Our findings offer relevant practical guidelines for GenAI to improve users' understanding of how to manage misinformation and how misinformation-processing influences their heuristic and procedural knowledge of it. Because diagnosticity is significantly influenced by the normative heuristics of AI systems, practitioners should not only establish the normative values of the system but also adopt strategies to increase the perceived heuristics of the diagnosticity of users. Examining both the presented information and user traits is critical for practitioners evaluating suitable venues for combating misinformation. In particular, because the credibility of information on AI platforms is low, users' distrust of such platforms may result in disinterest in discerning misinformation. Firms specializing in GenAI may invest resources and develop intervening features (e.g. explanatory cues) to monitor misinformation on their platforms, offer valuable cues or links to help users check or evaluate the credibility of information online, and engage in ethical and responsible operations to meet users' expectations of trust. Trust in a platform is generated when diagnosticity and established information credibility are high. Thus, to counter misinformation, practitioners should reveal the sources that share misinformation and relate it to the source instead of the platform on which the information was presented. Since this study showed the effectiveness of using explanatory interventions to reduce the effects of misinformation, this information sharing can be done through explanatory interventions.

When countering misinformation, it is also critical to nurture the literacy and efficacy of users. Diagnosticity may depend on the technological features of an AI system, but it can also be contextual, and the perceived diagnosticity of users can significantly hinge

on how they view the system's FAccT. Therefore, efforts should be made to improve the technical features and competence of systems for both diagnosticity and accuracy, and practitioners should establish the credibility of GenAI to enhance the literacy of users encountering misinformation.

Limitations and future research

Misinformation has become a complicated issue due to the emergence of GenAI and the growth of the new media landscape. We lay the groundwork for user misinformation models and present a user-driven approach to user misinformation analysis and characterization. Our study, like any other, has limitations that need to be addressed in future work; therefore, the findings should be considered with caution.

First, for parsimonious reasons, this study intentionally narrowed the applicability of the HS model, leaving out some of the critical components of the model, such as the characteristics of processing, conditions of heuristic use, and motivational assumptions (Todorov et al., 2002). The motivation to process AI-generated information carefully can also be related to contextual factors and factors related to time. For example, users may balance their need for a quick supply of information with their need for accuracy. The risk-to-benefit of finding accurate information can depend on the motivation to achieve accuracy. Because this study overlooked this factor, the applicability of the HS model was limited.

Second, it is worth noting that this study was conducted during the inception of GenAI. Thus, it is likely that the technology will develop and continue to expand as it advances. Many GenAIs, including ChatGPT, are in the early stages of development, and many more variations and possibilities will come. Although the affordances examined in this study solely include those with which we are currently familiar, AI's potential growth will go beyond what is now possible. Research on GenAI can be furthered to keep up with challenges and risks, including both technological and user-related affordances, before it achieves widespread adoption.

Third, although our study proposes a misinformation-processing model in the context of GenAI, it primarily addresses cognitive processing within an emerging single-user schema and excludes the algorithmic side of automatic processing. Future research should include the algorithmic features of information processing when using the model to more comprehensively test it. Further comparisons and investigations should be conducted regarding how and to what extent the different dimensions in the misinformation-processing model are shaped by the HS processing of users and how they influence the perceived diagnosticity of misinformation and the subsequent behaviors of users in turn. In addition, because we collected data at a single point in time, future studies should examine long-term exposure to misinformation and its longitudinal effects on people with longer and deeper engagement.

Fourth, prior research on misinformation processing has identified the role of trust, as people appear to be more accepting of information when they trust the media that presents it (Chen and Cheng, 2020). Users' existing trust in AI may play a significant role in the processing of misinformation. Future research should integrate diverse messages with varying levels of initial trust and investigate how prior trust levels may influence how users process misinformation. In doing so, misinformation processing may be

linked to a broader trust dimension, as mentioned in the existing literature (e.g. Stecula et al., 2020).

Finally, our results show a connection between FAccT, diagnosticity, and credibility, shedding light on the nexus between misinformation, literacy, and trust. These results suggest a possible two-tier or multiple-tier process that may play a role in users' susceptibility to misinformation. Future studies should test the relationships among algorithm literacy, trust, and diagnosticity to further explicate what may determine the efficacy and capacity of users when exposed to misinformation through algorithms. This task would warrant refining the two-tier process of misinformation.

Despite these limitations, our study offers meaningful insights into theory building and practice. The model offers a relevant framework for scholars interested in the behaviors of users who have been exposed to misinformation in a GenAI landscape. The model also provides the basis for future studies on cognitive heuristics that counter misinformation. Future research could combine algorithmic technologies and nudges in user experiences with GenAI services. The limitations of this study call for a more robust theoretical and empirical approach to the underlying processes of misinformation behavior in various contexts by considering the effects of personal traits on these processes.

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ORCID iDs

Donghee Shin  <https://orcid.org/0000-0002-5439-4493>

Joon Soo Lim  <https://orcid.org/0000-0003-0519-4169>

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Author biographies

Donghee Shin is Professor and Department Chair at Texas Tech University. Prior to Texas Tech, he worked as the head and director of the Interaction Science Center at Sungkyunkwan University. He was awarded an Endowed Chair Professorship by the Ministry of Education in Korea and Samsung Endowed Chair and served as Regent Professor at SKK University from 2011 to 2016.

Amy Koerber is Professor in Communication Studies and Associate Dean at Texas Tech University. Her most recent book, *“The Predatory Paradox: Ethics, Politics, and Practices in Contemporary Scholarly Publishing”* was published by Open Book Publisher in 2023. Dr. Koerber is PI on a recently awarded National Science Foundation grant, STEM Training in Ethics of Publication Practices.

Joon Soo Lim is Associate Professor of Public Relations at Newhouse School of Public Communications at Syracuse University. Dr Lim has been named a Knight News Innovation Fellow by the Tow Center at Columbia University. He was also named Page and Johnson Legacy Scholar from Arthur Page Center three times in 2012, 2017 and 2023 for his research on CSR communication and responsible use of digital analytics.

Appendix I. Measurements: A 7-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*).

Role	Variables	Measures	Source
Algorithmic heuristics	Fairness	1. GenAI shows no favoritism and does not discriminate against people (nondiscrimination). 2. The data sources from GenAI should be identified, logged, and benchmarked (accuracy). 3. I believe that GenAI follows the due process of impartiality with no prejudice (due process).	Mhasawade et al., 2021; Shin, 2023; Starke et al., 2023
	Accountability	1. I think that the system requires a person in charge who should be accountable for its adverse individual or societal effects in a timely fashion (responsibility). 2. Algorithms should be designed to enable third parties to examine and review the behavior of an algorithm (auditability). 3. Algorithms should have the option to modify a system in its whole configuration using only certain manipulations (controllability).	Barony & Reich, 2022; Shin, 2023
	Transparency	1. I think that the evaluation and the criteria of GenAI should be publicly released and understandable to people (understandability). 2. Any outputs produced by a GenAI system should be explainable to the people affected by those outputs (explainability). 3. GenAI should let people know how well the internal states of algorithms can be understood due to knowledge of its external outputs (observability).	Diakopoulos & Koliska, 2017; Molina & Sundar, 2023; Gran et al., 2021
Algorithmic moderator	Explainability	1. I find that GenAIs are easily understandable. 2. I think that GenAI services are interpretable.	Rai, 2020; Shin, 2023
	Accuracy	1. I can figure out the internal mechanics of GenAI. I hope that GenAI can be clearly explained. 2. In general, the items that GenAI systems recommend are precise. 3. GenAI-enabled recommendations are exact and correct.	Islam et al., 2020; Epstein et al., 2021; Pennycook & Rand, 2022; Lee et al., 2023; Peng & Wang, 2023
	Credibility	1. I trust GenAI's recommendations. 2. The items recommended through generative processes by AI are trustworthy. 3. I believe that the results of GenAI services are reliable.	Wu et al., 2022 Zhao & Tsang, 2020; Vu & Chen, 2023
Mediator	Diagnosticity	1. I expect this GenAI will help me get a real feel for the generated information. 2. I expect this GenAI to help me carefully evaluate this generated information. 3. Being able to get a real feel for a product would make it much easier for me to accept the information generated from this GenAI.	Chen & Cheng, 2020; Kim et al., 2020; Walter et al., 2021
Dependent variable	Intention to discern	1. I am largely pleased with GenAI services and intend to accept. 2. Overall, GenAI services fulfill my initial expectations and will use it. 3. I am generally happy with the content of GenAI services and will evaluate further.	Hwang & Jeong, 2021; Tully et al., 2020; Stecula et al., 2020; Lee et al., 2023